

2.1 Urbanisation and Urban Change

1. Global Urbanisation Trends

Urbanisation: The process by which an **increasing proportion** of a country's population comes to live in urban areas (towns and cities). It is measured as the percentage of total population living in urban areas — NOT the total number of people in cities.

Key Global Statistics

Year	% Urban Population (global)	Urban Population (approx.)
1950	30%	750 million
2007	50%	3.3 billion (first time majority urban)
2018	55%	4.2 billion
2050 (projected)	68%	~6.7 billion

Urbanisation rates vary dramatically by development level:

Region	% Urban (2020)	Trend
North America	82%	Slow growth; some counter-urbanisation
Europe	75%	Slow growth; some re-urbanisation (gentrification)

Latin America & Caribbean	81%	Largely already urbanised; growth slowing
Asia	51%	Rapid urbanisation ongoing — China, India, SE Asia
Africa	43%	Fastest urbanisation globally — doubling expected by 2050

Megacities

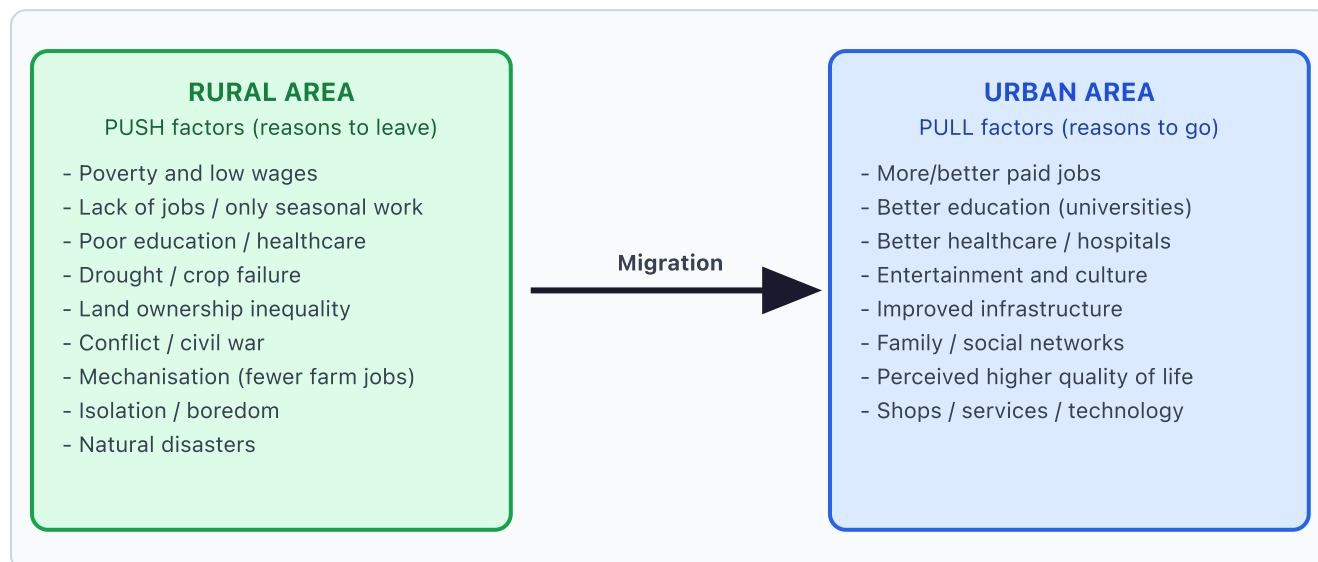
Megacity: A city with a population exceeding **10 million people**. In 1990 there were 10 megacities; by 2023 there are 35+. Most growth in megacities now occurs in LICs and NEEs.

City	Country	Population (urban agglomeration)	Classification
Tokyo	Japan	~37.4 million	HIC — world's largest city
Delhi	India	~32.9 million	NEE — fastest growing megacity
Shanghai	China	~29 million	NEE
São Paulo	Brazil	~22.4 million	NEE
Mumbai	India	~21 million	NEE
Lagos	Nigeria	~15 million	LIC — Africa's largest city
London	UK	~9.5 million	HIC

Exam Tip: The distinction between urbanisation and urban growth is a classic exam trap. Urbanisation = increasing *proportion* living in cities. Urban growth = increasing *number*. A city can grow in absolute population without urbanisation occurring if the rural population grows at the same rate. State both clearly in high-mark answers.

2. Causes of Urbanisation

Urbanisation results from two processes: **rural-urban migration** (people moving from countryside to city) and **natural increase** (birth rate exceeding death rate within cities).



Push-Pull Factors in Depth — With Statistics

Push-pull theory (Ravenstein, 1885) explains migration as driven by negative conditions at origin and positive conditions at destination. In reality, migration decisions are complex and shaped by personal networks, distance, cost, and government policies.

Push Factor	Supporting Evidence
Agricultural mechanisation reducing farm jobs	In India, agricultural employment fell from 60% of the workforce in 2000 to ~42% by 2020 (World Bank), displacing hundreds of millions from rural livelihoods.
Land inequality / landlessness	In Brazil in the 1970s–1990s, the top 1% of landowners held over 44% of agricultural land, pushing landless rural workers towards cities like São Paulo.
Climate shocks and drought	The 2009–2011 drought in the Sahel affected 10 million people; climate change could create 216 million internal climate migrants globally by 2050, 86 million of them in sub-Saharan Africa (World Bank, 2021).

Low rural wages	In Nigeria (2020), average rural income was approximately ₦30,000/month (~\$70) vs. urban minimum wage of ₦30,000 — but urban informal workers could earn 2–4x rural agricultural wages.
Conflict and instability	Syria's civil war (from 2011) displaced 6 million internally; many fled to Damascus, Aleppo, and eventually abroad. Northeast Nigeria's Boko Haram insurgency has driven millions into Lagos and southern cities.

Pull Factor	Supporting Evidence
Higher urban wages	The urban wage premium in developing countries is typically 15–30% above rural wages for equivalent work, and up to 200% for skilled urban jobs (ILO data).
Better education access	In Nigeria, urban areas have ~3x more secondary schools per capita than rural areas; university access is almost exclusively urban.
Healthcare	Under-5 mortality in rural Nigeria is 145/1,000 births vs. 88/1,000 in urban areas (NDHS, 2018) — a powerful pull for young families.
Social networks / family chain migration	Migrants follow established networks — a family member or friend who has already migrated reduces the cost, risk, and uncertainty of moving. This explains why migration streams often link specific rural regions to specific cities.
Perceived opportunity	Even the informal economy in Lagos or Mumbai offers more diverse income opportunities than subsistence farming — hawking, domestic work, construction, and petty trade all provide income that is irregular but potentially higher than rural work.

Exam Tip: When asked about push-pull factors, avoid listing them without context. Examiners at higher grades reward analysis: explain *why* a factor is significant (e.g. "Mechanisation is particularly powerful because it permanently removes jobs, leaving no reason to stay — unlike drought, which is temporary"). Use the word "because" repeatedly to force explanation rather than description.

Natural Increase in Cities

In LICs and NEEs, cities also grow through **natural increase** — a higher birth rate than death rate within the urban population. Migrants are often young adults (reproductive age), which raises the urban birth rate. Improved healthcare in cities reduces child mortality compared to rural areas. In Lagos, for example, the natural increase rate is approximately 3% per year — meaning even without migration, the city would double in population within 23 years.

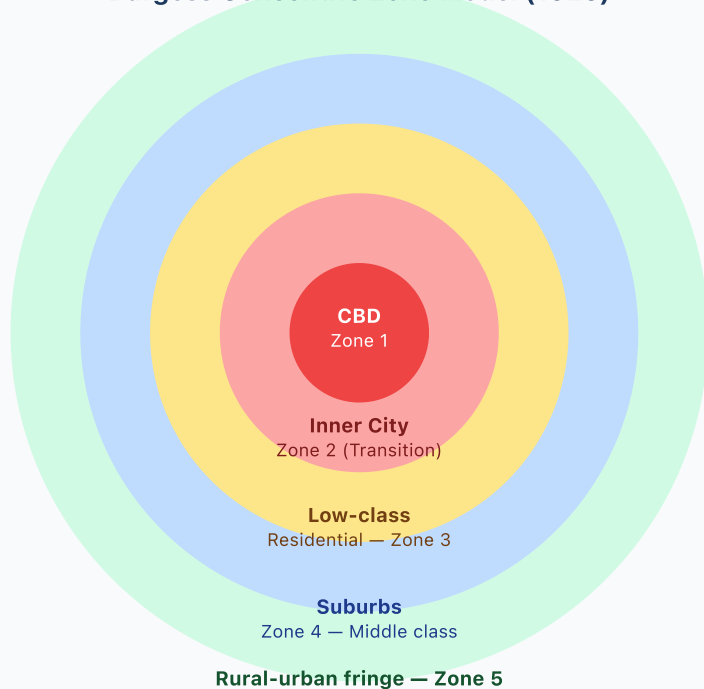
HIC vs LIC/NEE Urbanisation

Feature	HICs (e.g. UK, USA)	LICs/NEEs (e.g. India, Nigeria)
Stage of urbanisation	Already highly urbanised (75–85%); growth slowing or reversing	Rapidly urbanising; some cities growing at 3–5% per year
Main driver	Economic restructuring; suburbanisation/counter-urbanisation now prevalent	Rural-urban migration + high natural increase
Key challenge	Urban decay, deprivation, housing affordability, congestion	Insufficient housing, services, infrastructure for rapid population growth
Squatter settlements?	Rare / managed	Very common — ~1 billion people globally live in informal settlements
Historical timing	Urbanisation occurred gradually over 150+ years (Industrial Revolution from 1760s) allowing infrastructure to develop in parallel	Rapid urbanisation compressed into decades — infrastructure cannot keep up
Government capacity	Strong planning systems, building regulations, welfare provision	Often insufficient revenue, corruption, weak enforcement of planning law

3. Urban Land Use Models

Urban Land Use Model: A simplified diagram showing how land use (residential, commercial, industrial) is typically arranged within a city. Models are generalisations based on observations of many cities.

Burgess Concentric Zone Model (1925)



Burgess Concentric Zone Model

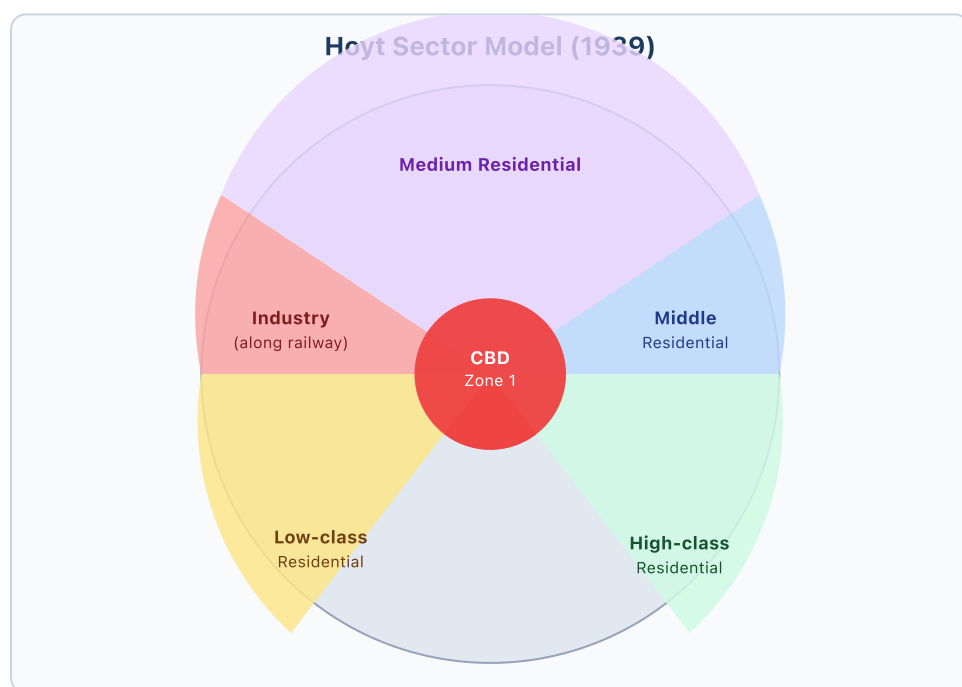
Developed by E.W. Burgess in 1925 based on Chicago. Proposes that cities grow **outward in concentric rings** from a central CBD. Land values decrease with distance from the centre; older housing is near the centre, newer suburbs on the edge.

Zone	Land Use	Characteristics
1 — CBD (Central Business District)	Shops, offices, entertainment, transport hubs	Highest land values; tallest buildings; maximum accessibility; little residential
2 — Zone of Transition (Inner City)	Industry, low-quality housing, warehousing	Old terraced housing, immigrants, overcrowding; transitioning to new uses; declining industry

3 — Low-class Residential	Workers' housing	Slightly better housing than Zone 2; older, smaller homes; working-class families
4 — Suburbs (Middle-class)	Residential — semi-detached housing, gardens	Lower density; better housing quality; car ownership; schools; quiet neighbourhoods
5 — Rural-Urban Fringe	Commuter housing, business parks, retail parks	Newest development; green belt pressure; out-of-town shopping; open space

Hoyt Sector Model (1939)

Homer Hoyt modified Burgess by showing that cities grow in **sectors (wedges)** along **transport routes** rather than perfect rings. His study of 142 American cities showed that land use does not radiate evenly but follows the lines of least resistance — roads, railways, and rivers.



Key features of the Hoyt Sector Model:

- Industry follows transport routes (rivers, railways) forming a wedge rather than a ring
- Low-class housing develops *adjacent* to industry (noise, pollution, cheap land)
- High-class housing develops on the **opposite side** of the city from industry, often on higher ground or near parks

- High-class sectors expand outward along desirable routes (e.g. suburbs along a scenic valley or arterial road)
- Explains pattern of many UK cities: e.g. West End (wealthy) vs East End (industrial/working-class) in London, or Headingley (affluent) vs East Leeds (industrial) in Leeds

Harris-Ullman Multiple Nuclei Model (1945)

Chauncy Harris and Edward Ullman proposed that cities do not develop around a single centre (CBD) but around **multiple nuclei** — separate centres that each attract certain land uses. This better reflects the reality of modern, complex cities.

- Certain activities cluster together (offices near offices; retail near retail)
- Some activities repel each other and cannot co-exist (heavy industry next to luxury housing)
- Some activities require specific sites (ports, universities, airports)
- High rents in the CBD push some activities to cheaper peripheral locations (e.g. out-of-town retail parks, science parks near universities)
- **Examples:** Edge cities in the USA (e.g. Tysons Corner, Virginia, near Washington DC); out-of-town shopping (Bluewater, Kent); business parks near airports; university campuses becoming economic nuclei

Exam Tip — Comparing Land Use Models: If asked to evaluate urban land use models, a strong answer acknowledges what each model explains well before criticising it. Burgess = good for showing land value gradients; Hoyt = better for explaining industrial/residential patterns; Harris-Ullman = most realistic for modern cities with edge cities and multiple retail/office clusters. Always link back to a real city example.

Limitations of Models

- All based primarily on American/European cities — may not apply to LIC cities
- Cities in LICs/NEEs often have the **poorest housing on the edge** (not the centre) as squatter settlements develop on cheap peripheral land
- Don't account for planning restrictions, motorways, historic town centres, edge cities
- Cities are becoming more fragmented and polycentric (multiple nuclei)

- Gentrification is reversing the original Burgess model in many HICs — inner city areas are now the most expensive (e.g. Shoreditch in London)

4. Urbanisation in LICs/NEEs — Squatter Settlements

Squatter settlement / Informal settlement / Slum: A densely populated area of a city where housing has been **illegally built** on land not owned by the residents, using temporary materials, without official permission. Known by various local names: *favela* (Brazil), *barrio* (Latin America), *shanty town* (general), *bidonville* (French Africa), *bustee* (India).

Why Do Squatter Settlements Develop?

- Rural-urban migrants arrive in cities faster than formal housing can be built
- Migrants cannot afford formal sector rents or house prices
- Land on the city periphery, flood plains, steep hillsides, or near railways/factories is undesirable — squatters occupy it with less risk of eviction
- Governments in LICs lack resources to provide social housing at scale
- Informal settlements develop incrementally — start with temporary shelters, gradually improve over years

Characteristics of Squatter Settlements

Feature	Detail
Housing	Built from salvaged materials — corrugated iron, plastic sheets, cardboard, breeze blocks. Often 1–2 rooms for extended families.
Water supply	Often no piped water; residents buy from water vendors at 10× the price paid by formal city residents, or use shared/contaminated sources
Sanitation	Few or no toilets/sewers; open sewers or defecation in the open; very high disease risk (cholera, typhoid, diarrhoea)
Electricity	Often illegally tapped from nearby power lines; fire risk; unreliable

Employment	Mostly informal economy — street vending, waste picking, domestic work, small manufacturing
Security	No legal title to land; constant threat of eviction; high crime rates in some areas
Health	High rates of respiratory disease (overcrowding, cooking fires); waterborne diseases; malnutrition; high infant mortality
Global scale	Approximately 1 billion people (1 in 7 globally) live in informal settlements (UN Habitat)

5. Case Study: Mumbai, India (NEE) — Squatter Settlements and Improvement

Case Study: Mumbai (NEE) — Urbanisation, Slums, and Improvement

Location and status: Mumbai (formerly Bombay) is India's financial capital, located on the west coast. It is India's largest city with ~21 million people in the urban agglomeration and ~24 million in the Greater Mumbai Metropolitan Region. It is one of the world's most densely populated cities. Approximately 60% of Mumbai's population lives in informal settlements occupying only 8% of the city's land area.

Why it grew so rapidly: Mumbai was the centre of British colonial trade (textiles, finance); post-independence it became India's financial and entertainment (Bollywood) hub. Millions migrated from rural Maharashtra, Uttar Pradesh, and Bihar seeking work and better wages.

Dharavi — Key Data: Located in central Mumbai, Dharavi covers approximately 2.1 km² and is home to an estimated 600,000–1 million people — making it one of Asia's most densely populated areas and one of its largest informal settlements.

- **Population density:** Up to 300,000 people per km² — approximately 10 times the density of Manhattan, New York
- **Sanitation crisis:** 1 toilet per ~1,440 residents (compared to UN standard of 1 per 50)
- **Living space:** Most homes are 10–15 m² housing 5–10 people
- **Average income:** Approximately \$500 per year — well below Mumbai's average of ~\$3,000

- **Child health:** Infant mortality in Dharavi is approximately 68 per 1,000 live births vs. 36 per 1,000 for Mumbai as a whole
- **Education:** An estimated 25% of children in Dharavi do not attend school regularly; many work in family businesses

Dharavi's informal economy: Despite its challenges, Dharavi is a productive economic zone with an estimated annual turnover of **\$600–700 million USD**. It contains ~15,000 single-room factories and 5,000 businesses including:

- **Recycling:** Processes up to 80% of Mumbai's plastic waste; workers (called "kabariwalas") sort and sell recyclables
- **Leather goods:** Many bags and accessories sold globally are produced in Dharavi workshops
- **Pottery (Kumbharwada):** Traditional pottery quarter has operated for over 100 years
- **Food production and garment manufacturing**

This demonstrates that informal settlements are not simply places of poverty — they are functional communities with complex economies.

Dharavi Redevelopment Project: A proposed redevelopment (approved in principle 2022, led by the Adani Group) aims to rehouse all residents in modern apartment blocks on the same site, while building ~76,000 additional market-rate apartments to fund the project. Critics argue: community businesses and social networks will be destroyed; many residents will not qualify for free housing due to technicalities; high-rise living may not suit residents' occupational needs (potters, leather workers need ground-floor space). Supporters say it will deliver piped water, sewage, electricity, and schools to all residents.

Squatter Settlement Improvement Schemes

Three main approaches are used by governments and NGOs to improve informal settlements:

Approach	How it works	Advantages	Disadvantages
Self-Help Schemes (SHS)	Government provides materials (bricks, cement, roofing), legal land tenure, loans,	Cheaper per family than government-built housing; community ownership and	Quality depends on residents' skills; can be slow; corruption in allocation; may exclude the

	and technical advice; residents build/improve homes themselves	investment; preserves social networks; houses tailored to residents' needs	poorest who cannot contribute labour
Site and Service Schemes	Government provides serviced plots (road access, basic water supply, sewerage connection) and residents build their own home on the plot	More affordable than completed housing; residents invest in and maintain their own property; builds permanent community; used widely in East Africa and South Asia	Residents still bear building costs; initial site may lack full services; risk of subletting or resale to higher-income groups ("elite capture")
Slum Clearance / Forced Relocation	Government demolishes informal settlement and rehouses residents in purpose-built blocks elsewhere in the city	Can deliver high-quality permanent housing quickly; clears land for infrastructure development; improves area statistics	Destroys community and economic networks; new housing often far from residents' workplaces; high-rise flats unsuitable for families with outdoor businesses; residents may return to new slums; social trauma; often used to clear land for commercial development rather than residents' benefit

Mumbai — Slum Rehabilitation Authority (SRA) Data:

- Established 1995; has rehoused an estimated 600,000+ slum dwellers in permanent buildings under the Slum Rehabilitation Scheme
- The scheme provides free 269 sq ft (25 m²) apartments to qualifying slum dwellers; developers are rewarded with additional "sale component" apartments for the market
- By 2023, approximately 30,000 buildings had been transferred to slum dwellers — but Mumbai still has ~2,000 notified slum pockets, home to an estimated 5–6 million people

- **Criticism:** Many residents are excluded for not meeting the "cutoff date" criteria (must have lived there before 1 January 1995 to qualify); the 25 m² apartments are considered too small for families; buildings have had maintenance problems and lift failures

Exam Tip — Squatter Settlement Questions: Exam questions often ask you to "evaluate" improvement schemes. Always discuss at least two approaches and weigh pros vs. cons. The key evaluative point is that self-help and site-and-service schemes respect community agency, while slum clearance often serves developer or government interests more than residents. Support with specific Dharavi or Mumbai SRA data for top marks.

6. Case Study: Lagos, Nigeria (LIC) — Urbanisation in Africa

Case Study: Lagos, Nigeria — Africa's Megacity

Background and Location: Lagos is Nigeria's largest city, located on the Bight of Benin (Atlantic coast) in southwest Nigeria. It is the economic capital of Nigeria (though Abuja is the political capital) and the largest city in sub-Saharan Africa, with a metropolitan population of approximately 14–15 million (2023), growing at around 3% per year. By 2050, the UN projects Lagos will have over 30 million people.

Population Growth Timeline:

- 1950: ~290,000 people
- 1970: ~1.4 million (Lagos was Nigeria's capital and a major port)
- 1990: ~5 million (rapid industrialisation; oil economy attracting workers)
- 2000: ~7.5 million
- 2015: ~12 million
- 2023: ~15 million (metropolitan area); ~24 million (urban agglomeration including surrounding areas)

Why Lagos grew so fast:

- Nigeria's oil boom from the 1970s concentrated wealth and government spending in Lagos
- Colonial legacy: Lagos was Nigeria's administrative and commercial centre under British rule

- Rural push from agricultural collapse: Nigeria went from a food exporter in 1960 to an importer by the 1980s; the oil boom destroyed agricultural competitiveness, pushing rural workers to cities
- Civil war (Biafra, 1967–1970) displaced millions; many eventually settled in Lagos
- High natural increase: Nigeria has a total fertility rate of ~5.3 children per woman (2020); urban birth rates remain high
- Informal economy opportunities: Lagos generates approximately 25–30% of Nigeria's total GDP despite having only ~8% of the population

Infrastructure Challenges:

- **Traffic:** Lagos is consistently ranked among the world's worst cities for traffic congestion (TomTom Traffic Index). Average commute times exceed 2 hours each way. The city's road network was designed for ~1 million people. The Lagos BRT (Bus Rapid Transit), opened 2008, carries 200,000+ passengers/day — Africa's largest BRT — but is overwhelmed.
- **Water:** Lagos Water Corporation officially supplies 210 million litres/day — but the city needs 700+ million litres/day. Most residents rely on private boreholes (often contaminated) or buy from water vendors at up to 10× the piped water price.
- **Electricity:** EKEDC (Lagos electricity distributor) provides power for an average of 8–12 hours/day in better-served areas, often far less. Most businesses run on diesel generators, adding to air pollution and costs.
- **Waste:** Lagos produces approximately 10,000 tonnes of solid waste per day; formal collection capacity is ~40% of this. The Olusosun landfill (one of the world's largest) has received waste for 30+ years and is now critically over capacity.
- **Flooding:** Much of Lagos is built on low-lying islands and coastal land. Annual flooding affects millions of residents; climate change is increasing flood frequency and severity.

Makoko — The Floating Slum: Makoko is one of Lagos's most well-known informal settlements, built on stilts over Lagos Lagoon. Key facts:

- Population estimated at 100,000–300,000 people (difficult to census accurately)
- Buildings are constructed on wooden stilts above the water; residents travel by canoe
- No formal sanitation — sewage goes directly into the lagoon; the water is heavily polluted
- No piped water; residents use lagoon water (contaminated) or buy bottled/tanker water

- Children are rowed to school in canoes; there is a floating school (though the original structure collapsed in 2016 and a new one was rebuilt)
- Community is mostly from the Egun fishing community who have lived on the lagoon for generations — it is both an economic necessity and a cultural community
- In 2012, Lagos state government partially demolished Makoko with minimal notice, triggering international protests from human rights organisations. Many residents rebuilt.
- **Why it exists:** Land on Lagos Island is among the most expensive in Africa; waterfront land is formally controlled. The lagoon is unregulated space where poor migrants can build without land costs.

The Informal Economy in Lagos:

- An estimated 65–70% of Lagos workers are in the informal economy — street hawking, small trade, motorcycle taxis ("okadas"), domestic work, building and repair
- "Alaba International Market" — the world's largest electronics market — is largely informal
- The Lagos informal economy generates an estimated \$35 billion USD per year (McKinsey, 2014)
- Okada (motorcycle taxis) provided critical transport for millions of poorer residents — the Lagos state government banned them from major roads in 2022, displacing hundreds of thousands of workers overnight, illustrating the fragility of informal livelihoods

Improvements and Government Response:

- Lagos BRT (2008) — Africa's largest Bus Rapid Transit system, 22 km route, 200,000+ passengers/day
- Eko Atlantic — a new city being built on reclaimed land adjacent to Victoria Island, designed for businesses and wealthy residents. Critics call it a "city for the rich" that does nothing for the urban poor.
- Lagos State has invested in CCTV, policing reform, and formal markets to reduce crime
- Drainage improvement projects to reduce flooding — though progress is slow relative to population growth

Exam Tip — Lagos vs Mumbai: Lagos (LIC) and Mumbai (NEE) are often compared. Key contrasts: Nigeria's GDP per capita (~\$2,200) is similar to India's (~\$2,400), but India's urban

middle class is much larger. Lagos has less formal government capacity to fund improvement schemes than Mumbai. Lagos also faces the added challenge that Nigeria's federal government revenues are dominated by oil — which creates boom-bust cycles affecting infrastructure investment. In exam answers, specify whether a problem is LIC/NEE-specific or universal.

7. Urban Issues in LICs/NEEs

Issue	Detail	Example
Traffic congestion	Rapid motorisation as incomes rise; poor road infrastructure; mixed traffic (pedestrians, rickshaws, cars, buses); poor public transport	Mumbai: average commute speed 12 km/h in peak hours; Delhi: second most congested city globally (TomTom 2023); Lagos: commuters spend average 4+ hours in traffic daily
Air pollution	Vehicle exhaust (older, poorly maintained vehicles); industrial emissions; domestic cooking fires; construction dust	Delhi: PM2.5 levels regularly 10–20× WHO safe limit; "AQI hazardous" for weeks in winter; causes 1.7 million deaths in India annually. Lagos: vehicle emissions worsened by fuel quality (high-sulphur diesel)
Inadequate sanitation	Insufficient sewage infrastructure; open sewers; rivers used as open sewers	Dharavi: 1 toilet per 1,440 people; Mithi River in Mumbai severely polluted; Makoko (Lagos): sewage directly into lagoon
Water supply	Pipes inadequate; contamination; inequitable supply (rich areas get piped water, poor pay vendors); groundwater over-extraction	Lagos: formal supply only 30% of need; Mumbai gets water for only 2–4 hours/day in many areas; Jakarta sinking due to groundwater extraction
Waste management	Insufficient formal collection; illegal dumping; open landfills; burning of waste	Lagos: Olusosun landfill critically overcapacity; waste pickers (informal recyclers) handle much of the city's waste; Dharavi processes 80% of Mumbai's plastic waste informally
Crime and insecurity	Poverty, inequality, unemployment — especially youth unemployment	São Paulo: gang-related violence; Lagos: "area boys" gangs controlling markets; rapid decline

— drives property crime and gang activity

in murder rate in some favelas following targeted social programmes

8. Urbanisation in HICs — Counter-Urbanisation and UK Trends

In HICs, urbanisation has largely completed. Instead, three counter-trends are observed:

Process	Definition	Causes	Effects
Suburbanisation	Movement of people and businesses from the city centre to the outskirts	Mass car ownership (20th century); desire for space, gardens, better schools; white flight (racial); council estate building on city fringes	Urban sprawl; loss of agricultural land; decline of inner city; out-of-town retail; longer commutes
Counter-urbanisation	Movement of people away from cities to smaller towns, villages, and rural areas	Remote working (accelerated by COVID-19); desire for quality of life; internet connectivity in rural areas; retirement; lower house prices	Rural house price inflation; pricing out of locals; loss of rural character; commuter villages; rural gentrification
Re-urbanisation	Return of people (often young professionals) to city centres	Regeneration of inner-city areas (Docklands, Salford Quays); gentrification; desire for walkable, cultural urban living; delayed car ownership among millennials	Gentrification displaces existing communities; housing prices rise; cultural homogenisation; loss of affordable housing

Counter-Urbanisation in the UK — Detailed Analysis

Counter-urbanisation in the UK has been ongoing since the 1970s but received a dramatic acceleration during and after the COVID-19 pandemic (2020–2022). Key data and trends:

COVID-19 and Counter-Urbanisation (2020–2022)

- During the 2020 lockdowns, approximately **700,000 people left London** (according to Royal Mail address data and GP registrations), many permanently relocating to rural and coastal areas
- Rightmove (UK property search website) reported a **126% increase in searches for rural properties** in summer 2020 vs. summer 2019
- The "race for space" saw demand for detached houses with gardens surge dramatically, while city-centre flats declined in relative value
- Popularised areas: Cornwall, Devon, Dorset, Norfolk, Northumberland, Yorkshire Dales — all saw significant in-migration from cities
- Office for National Statistics (ONS) data showed that by 2021, remote working was being done by 38% of UK workers at least some of the time — up from 12% pre-pandemic — removing the need to live close to the workplace

Rural House Price Inflation

- Average house prices in Cornwall rose by **20% in 2020 alone** (Savills estate agents), the fastest growth of any English county
- In mid-Wales and the Yorkshire Dales, some villages saw house prices double between 2019 and 2022
- The average house price in many rural English villages now exceeds £400,000 (ONS, 2023), while average rural salaries are significantly below urban equivalents
- The ratio of house prices to local earnings in rural areas rose from approximately 8:1 in 2010 to over 12:1 in 2022 — making it impossible for local young people, agricultural workers, and key workers to buy homes in their communities

Impacts on Rural Communities

Impact	Detail
Loss of affordable housing	Local first-time buyers priced out; housing associations unable to compete with private buyers; young people forced to leave the communities where they grew up
Second homes problem	Cornwall: approximately 14,000 registered second homes (2022); many villages (e.g. Rock, Padstow) have majority second-home ownership — meaning communities "die" in winter when owners return to cities

Services under pressure	Influx of higher-income residents raises demand for services but may not increase year-round viability of local schools, GP surgeries, or post offices — these depend on permanent residents
Cultural change	Traditional rural occupations (farming, fishing) become economically marginal as land and housing costs rise for workers; rural character changes
Political responses	Wales introduced a 300% council tax surcharge on second homes (2023); some areas in England have introduced Article 4 Directions restricting conversion of homes to holiday lets without planning permission

Exam Tip — Counter-Urbanisation Questions: Always specify a time period for UK counter-urbanisation data. Pre-2020 data shows gradual movement; post-2020 data shows dramatic COVID-19 acceleration. The strongest exam answers discuss both push factors from cities (high costs, pandemic fears, cramped flats) and pull factors to rural areas (space, remote working, "quality of life" perception) — and then evaluate the negative consequences for rural communities, particularly housing affordability. Include at least one statistic.

Gentrification: The process by which an area of lower-income, working-class housing is gradually taken over by higher-income, middle-class residents and businesses, raising property values and displacing existing residents who can no longer afford to live there. Example: Brixton (London), Hackney (London), Northern Quarter (Manchester).

Urban Sprawl: The unplanned, uncontrolled spread of urban areas into surrounding rural land. Driven by suburbanisation and out-of-town development. The UK Green Belt policy (1955) was designed to prevent urban sprawl around major cities.

9. Sustainable Cities — LICs, NEEs, and HICs

Sustainable city / sustainable urban development: A city that meets the needs of its current residents without compromising the ability of future generations to meet their needs. Involves

balancing economic prosperity, social equity, and environmental protection. The UN Sustainable Development Goal 11 calls for "sustainable cities and communities" by 2030.

Sustainability in urban areas involves:

- **Environmental sustainability:** Reducing carbon emissions, managing waste and water, protecting green space, using renewable energy
- **Social sustainability:** Providing affordable housing, equitable access to services, reducing inequality, inclusive communities
- **Economic sustainability:** Creating stable, diverse economies that provide livelihoods for all residents, not just the wealthy

Case Study: Curitiba, Brazil — Sustainable Urban Planning in an NEE

Case Study: Curitiba, Brazil — A Model of Sustainable Urban Planning

Background: Curitiba is the capital of Paraná state in southern Brazil, with approximately 1.9 million people in the city and 3.6 million in the metropolitan area (2023). Like many Brazilian cities, it experienced rapid population growth in the 20th century — its population grew 10-fold between 1950 and 2000. However, under Mayor Jaime Lerner (architect and urban planner, mayor 1971–1975, 1979–1983, 1989–1993), Curitiba adopted an integrated urban planning approach that became internationally celebrated.

Key Sustainable Features:

1. Bus Rapid Transit (BRT) System:

- Curitiba pioneered the modern BRT system from the 1970s — a network of dedicated bus lanes with tube-shaped stations where passengers pay before boarding, allowing rapid loading
- The system carries approximately **2 million passengers per day** — more than the population of the city itself, because it serves the wider metropolitan area
- Buses run every 90 seconds on key routes; the system has over 370 km of bus lanes
- Car use in Curitiba is among the lowest of Brazilian cities despite high car ownership rates — residents choose to use the bus because it is fast, affordable, and reliable
- The Curitiba BRT model has been replicated in over 200 cities globally, including Lagos (Lagos BRT) and Bogotá (TransMilenio)

2. Green Space:

- Curitiba has over 52 m² of green space per inhabitant — the UN recommended minimum is 12 m². This is among the highest of any city in the world.
- The city converted flood-prone riverside land into linear parks rather than building on it — solving both flood management and green space provision simultaneously
- By 2020 the city had 30 parks and 1,000 green squares, up from just 2 parks in the 1970s

3. Waste Management and Recycling:

- Curitiba recycles approximately **70% of its household waste** — one of the highest rates in the Americas
- The innovative "Green Exchange" programme allows low-income residents (particularly in informal settlements) to exchange sorted recyclable waste for food vouchers, bus tokens, or school supplies — incentivising recycling while addressing poverty
- By 2020, the programme had exchanged 11,000 tonnes of recyclables for 1.2 million food baskets, providing food security while managing waste

4. Affordable Housing and Social Integration:

- Curitiba built affordable housing along BRT corridors rather than in peripheral locations — ensuring low-income residents had good transport access to jobs
- Land regularisation schemes gave informal settlement residents legal title to their plots, enabling investment in their homes

Criticism and Limitations:

- Curitiba still has significant favelas (called "vilas" locally) — by 2023 approximately 15% of residents live in informal settlements
- The city's success has attracted in-migration, putting pressure on affordable housing
- The BRT system is overcrowded at peak times; some analysts argue a light rail system is now needed
- Curitiba's model worked partly because Jaime Lerner operated with strong mayoral authority over long periods — replicating this in cities with different political systems is difficult
- Some critics argue Curitiba's reputation has been overstated — Brazilian cities with less planning attention (e.g. Porto Alegre) have similar or better outcomes on some measures

Why Curitiba Matters for Exams: It demonstrates that sustainable urban development is achievable in an NEE city with limited resources if there is strong political will and integrated planning. The BRT system is the most widely cited example of LIC/NEE urban sustainability globally.

Sustainability Strategies Across Development Levels

Strategy	HIC Example	LIC/NEE Example
Renewable energy	Copenhagen, Denmark: 62% of electricity from wind power; district heating network from waste heat	Curitiba, Brazil: bio-gas from landfill used for electricity generation
Sustainable transport	Amsterdam: 63% of trips made by bicycle; extensive tram and metro network	Curitiba: BRT carries 2 million/day; Lagos BRT 200,000/day
Waste recycling	Germany: 66% household recycling rate; strict EU waste regulations	Curitiba: 70% recycling rate; Green Exchange programme
Green space	Singapore: "City in a Garden" — 47% land area as green space; vertical gardens	Curitiba: 52 m ² green space per person; linear riverside parks
Affordable housing	Vienna, Austria: 60% of residents in subsidised housing; strong social housing model	Mumbai SRA: 600,000+ residents rehoused in permanent buildings
Waste water management	London: Thames Tideway Tunnel (super sewer) reducing raw sewage overflow	Dhaka, Bangladesh: Korail slum — NGO-led community water kiosks reducing waterborne disease

Exam Tip — Sustainable Cities Questions: When writing about sustainable cities, structure your answer around the three pillars: environmental, social, and economic sustainability — and make sure Curitiba addresses all three (BRT = economic and environmental; Green Exchange = social

and environmental; green space = environmental and social). Avoid describing Curitiba as "perfect" — acknowledging its limitations shows the higher-level thinking examiners reward.

10. Urban Functions and Hierarchy

Settlements can be arranged in a **hierarchy** based on population size and the range of services they provide. Higher-order settlements have more varied and specialised functions, a larger sphere of influence, and are further apart.

Settlement Type	Population	Functions / Services	Sphere of Influence
Hamlet	<100	None / very basic (pub, farm)	Very local (<1 km)
Village	100–2,000	Post office, primary school, church, local shop	Surrounding farms (~5 km)
Town	2,000–100,000	Secondary school, supermarket, cinema, hospital	Several surrounding villages (20–50 km)
City	100,000–1 million	University, regional hospital, major shopping, theatres	Regional (50–200 km)
Conurbation	1–10 million	Multiple CBDs, international transport hub, financial services	National / international
Megacity	>10 million	All major global functions; global financial markets; multiple universities	Global

Central Place Theory (Walter Christaller, 1933): Proposes that settlements are distributed across space in a regular, hierarchical pattern. **Threshold** = minimum population needed to support a service. **Range** = maximum distance people will travel for a service. Higher-order settlements (cities) have higher thresholds and ranges and are spaced further apart.

11. Evaluation — HIC vs LIC Urbanisation Challenges Compared

A strong GCSE answer recognises that urban challenges in HICs and LICs/NEEs are fundamentally different in scale, cause, and available solutions — but some challenges (inequality, housing affordability, pollution) are universal.

Challenge	HICs (e.g. UK cities)	LICs/NEEs (e.g. Lagos, Mumbai)	Key Difference
Housing affordability	London average house price ~£526,000 (2023); average salary ~£41,000; ratio ~12:1. Young people locked out of ownership. Driven by lack of supply and investment demand.	Mumbai: 60% of population in informal settlements on 8% of land. No option of formal housing — too expensive and insufficient supply. Driven by scale of in-migration and poverty.	HIC problem = price; LIC problem = existence of formal housing at all. HIC solutions include planning reform, rent control. LIC solutions include self-help schemes, which are irrelevant in HICs.
Traffic and congestion	London: congestion charge (£15/day into central London) has reduced traffic 25% since 2003; extensive Underground/bus network. Problem is manageable.	Lagos: average commute 4+ hours/day; road network designed for 1 million but serving 15+ million; Lagos BRT overwhelmed. Problem is structural and worsening.	HICs have financial capacity to invest in solutions; LICs cannot fund infrastructure at the speed needed.
Air pollution	UK NO ₂ levels in cities exceed WHO limits; major cause is diesel vehicles. London ULEZ (Ultra Low Emission Zone) has reduced NO _x by 44% in the zone since 2019.	Delhi: PM2.5 regularly 20× WHO limit; causes 1.7 million deaths/year in India. Driven by vehicle exhaust, industrial emissions, crop burning, and domestic cooking fires simultaneously.	LIC pollution is multi-source and harder to regulate; HIC pollution is reducing through regulation and vehicle technology.

Waste management	UK recycles 45% of household waste; regulated landfills; incineration with energy recovery. Problem is moving towards zero-waste economy.	Lagos: formal collection handles 40% of 10,000 tonnes/day generated; Olusosun landfill critically overcapacity. Growing faster than solutions.	Scale of the problem in LICs vastly exceeds government capacity; in HICs it is an environmental quality problem, not a public health emergency.
Social inequality	Gini coefficient UK: ~0.36 (moderate inequality); deprivation concentrated in specific inner-city or post-industrial areas (e.g. Middlesbrough, Blackpool). Managed through welfare state.	Gini coefficient Nigeria: ~0.43; Brazil: ~0.53 (high inequality). Informal settlement residents earn a fraction of wealthy residents; no universal welfare safety net; inequality is spatial and extreme.	Both have inequality but LIC inequality is more extreme, more spatially concentrated, and has fewer state mechanisms to address it.
Environmental sustainability	UK has legally binding net-zero carbon targets (2050 Climate Change Act); cities reducing carbon through EV charging, heat pump installation, building insulation schemes.	Curitiba has pioneered BRT and recycling; but Lagos, Kinshasa, Dhaka face overwhelming challenges of basic service provision before environmental sustainability can be prioritised.	Sustainability is a luxury of development; HICs can pursue it directly, LICs must combine sustainability with basic provision.

Exam Tip — Evaluation Questions (6–9 marks): The highest marks in evaluation questions go to answers that avoid simple "LICs are worse" statements and instead explain WHY differences exist. Key explanatory factors: (1) speed of urbanisation — HICs had 150 years to build infrastructure; LICs are growing 10× faster; (2) government revenue — HICs have tax bases to fund solutions; LICs do not; (3) scale — Mumbai's informal population alone (~12 million) exceeds the total population of many European countries. Always conclude with a justified overall judgement: e.g. "Overall, LIC urbanisation presents greater immediate challenges because the scale of in-

migration exceeds any government's capacity to respond, whereas HIC challenges, though serious, are more manageable with existing resources."

Key Distinction: Urbanisation = the PROCESS (increasing % urban). Urban growth = the increase in absolute NUMBER of people in cities. A city can grow in population without urbanisation occurring if the rural population grows equally fast. This distinction matters in exam answers.